Technical Service Bulletin

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QSK60 CM850 Modular Common Rail System Engine Introduction	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin introduces the Cummins® QSK60 Engine with CM850 Modular Common Rail System. The QSK60 is a 16 cylinder engine with a 60 liter displacement. The engine has been designed to meet United States Tier 2 mobile off-highway emission levels. The United States Tier 2 mobile off-highway emission requirements became effective January 1, 2006 for ratings above 560 kW (751 hp).

The upgraded design includes a new high pressure modular common rail fuel system and advanced ECM. New electronically actuated injectors, pistons, turbocharger, and valve covers have also been incorporated into the configuration. The QSK60 MCRS will continue to utilize a 2-pump, 2-loop low temperature aftercooling system to deliver the intake manifold temperatures that are required for Tier 2 emission levels.

General Engine Data	
QSK60 With CM850 Modular Common Rail System Engine Data	
Configuration	D593007CX03
Configuration	D593006CX03
Number of Cylinders	V - 16
Displacement	60 liter [3661 in ^{3}]
Bore	158.75 mm [6.25 in]
Stroke	190 mm [7.5 in]
Compression Ratio	14.5:1
Aspiration (Single Stage turbocharging)	Turbocharged and aftercooled
Aspiration (Two Stage turbocharging)	Turbocharged, intercooled, and aftercooled
Approximate engine weight (Single Stage turbocharging)	7800 kg [17,196 lb] (estimate dry)

General Engine Data	
QSK60 With CM850 Modular Common Rail System Engine Data	
Approximate engine weight (Two Stage turbocharging)	9000 kg [19,842 lb] (estimate dry)

Engine ratings available for the QSK60 with CM850 Modular Common Rail System engine are essentially ratings from the QSK60 mechanically actuated injector engine. Tier 2 performance curves offer the same torque rise as Tier 1 curves. The performance parts have been developed to meet Tier 2 emission levels, achieve the best possible fuel consumption, and achieve similar transient response as the QSK60 mechanically actuated injector engine.

Option Number	Rated Horsepower	Torque Peak	Altitude Capability (ft)	Turbocharger arrangement	Rating Type
FR6502	1400hp @ 1800 rpm	4710 lb-ft @ 1300 rpm	10,000	Single Stage	Continuous
FR6490	1500 hp @ 1900 rpm	5042 lb-ft @ 1300 rpm	6,000	Single Stage	Intermittent
FR6452	2000 hp @ 1900 rpm	5805 lb-ft @ 1500 rpm	8,000	Two Stage	Intermittent
FR6574	2250 hp @ 1900 rpm	6299 lb-ft @ 1500 rpm	4,500	Two Stage	Restricted
FR6545	2300 hp @ 1900 rpm	6512 lb-ft @ 1500 rpm	4,000	Two Stage	Restricted

Engine Dimensions	
Dimension Description	**Dimension**
Overall height of Single Stage engine above the centerline of crankshaft	1176 mm [46.3 in]
Overall height of Two Stage engine above the centerline of crankshaft	1524 mm [60.0 in]
Overall height below the centerline of crankshaft	789 mm [31.1 in]
Overall length from rear face of flywheel housing to fan drive mounting surface	2608 mm [102.7 in]
Overall width at widest point	1588 mm [62.5 in]

Training for the QSK60 with CM850 Modular Common Rail System (MCRS), control system is contained in Cummins® Virtual College training modules. This will be a part of the High Horsepower Virtual College.

The cooling system is a 2-pump, 2-loop system that requires both an engine coolant loop radiator and a low temperature aftercooling (LTA) radiator.

The total system heat rejection has increased slightly as a result of the revised combustion characteristics required to meet Tier 2 EPA emission levels. Factors that contribute to this are increased turbocharger compressor outlet temperatures and increased air flow. Maximum top tank temperature, however, remains unchanged at 100°C [212°F]. The service interval of the corrosion filter is still 250 hours.

One of the key differences in the cooling system when compared to the QSK60 mechanically actuated injector is the water pump. The QSK60 MCRS engine is equipped with a water pump that incorporates the impeller for the engine coolant loop and the low temperature aftercooler coolant loop on the same drive shaft. The inlet for both circuits is on the right-bank side of the engine.

The torque converter cooler option remains the same and the corrosion filter continues to be required and remains unchanged. The service interval of the corrosion filter is 250 hours.

The cooling system is driven by a water pump that has two impellers driven off of a common shaft. This water pump is located in the same place as the water pump on the QSK60 mechanically actuated injector engine. Each impeller drives one of the cooling system loops. There are 2 inlets on this water pump, one for each circuit. See Figure 1 below.

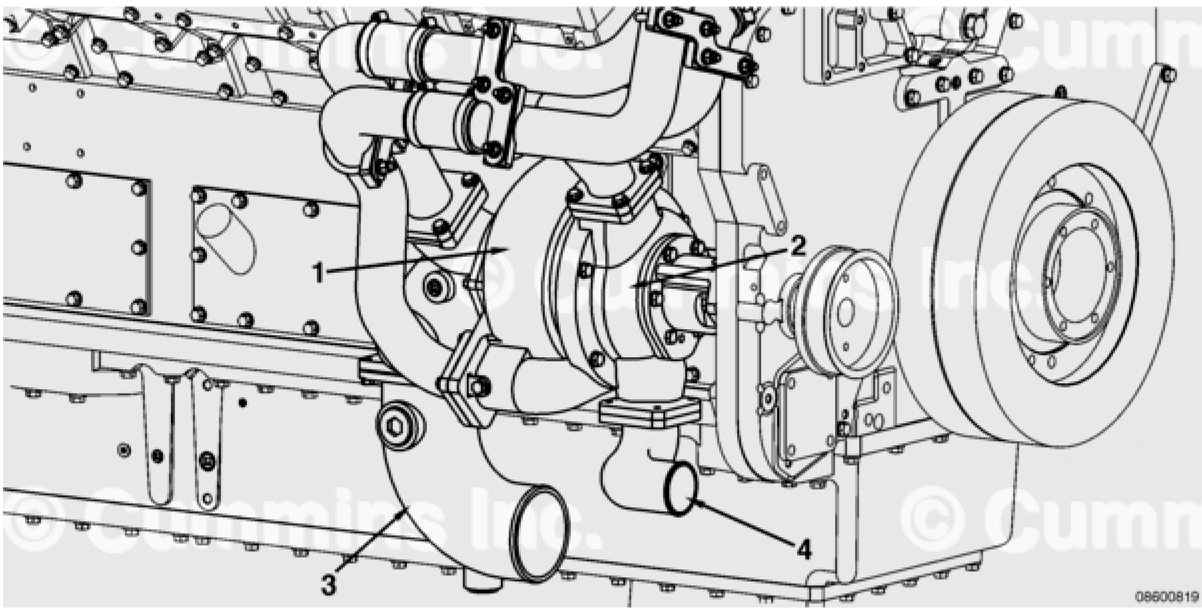

Figure 1, Water Pump

1. Engine coolant impeller
2. Low temperature aftercooler coolant impeller

3. Engine coolant inlet (shown with optional inlet connection)
4. Low temperature aftercooler coolant inlet (shown with optional inlet connection)

The low temperature aftercooler outlet connection and the engine coolant inlet and outlet connections are located in the same areas as the QSK60 mechanically actuated injector coolant connections. The connections have changed slightly. The low temperature aftercooler inlet connection has been moved to the water pump on the right bank of the engine. (See Figures 1 and 2). Please reference the installation drawing for more detailed information.

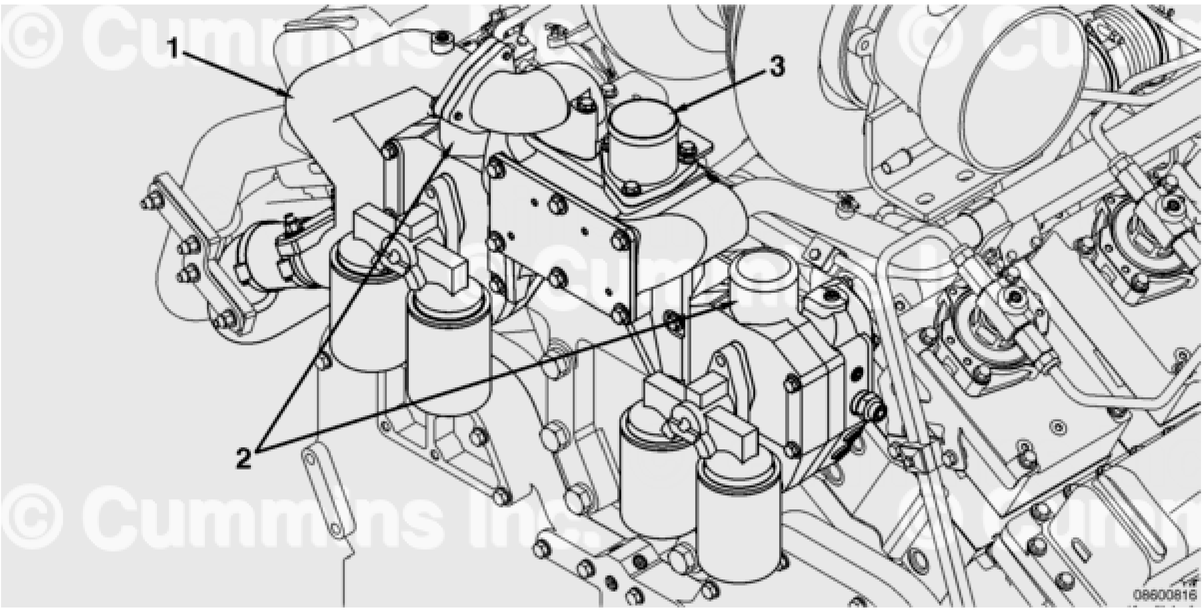

Figure 2, Inlet and Outlet Connections

1. Low temperature aftercooler bypass
2. High temperature engine coolant outlet
3. Low temperature aftercooler coolant outlet.

A new option has been released to allow diversion of coolant to a torque converter. Selecting this option adds a coolant supply connection to the water pump housing. Corrosion filters will continue to be required and remain unchanged. The filters are available on either the pan rail or the front of the thermostat housing. Cab heater supply connections are available at the rear of the water manifold. Cab heater return connections are available at the inlet of the water pump.

The 2-pump, 2-loop cooling system employed on the QSK60 MCRS results in a cooling system that is slightly different from the QSK60 mechanically actuated injector engine. As mentioned previously, the low temperature aftercoolers are located on their own loop. The main engine loop now consists of the oil cooler, engine block, jacket water thermostats, and water pump. The two cooling circuits share a common top tank. See Figure 3 below.

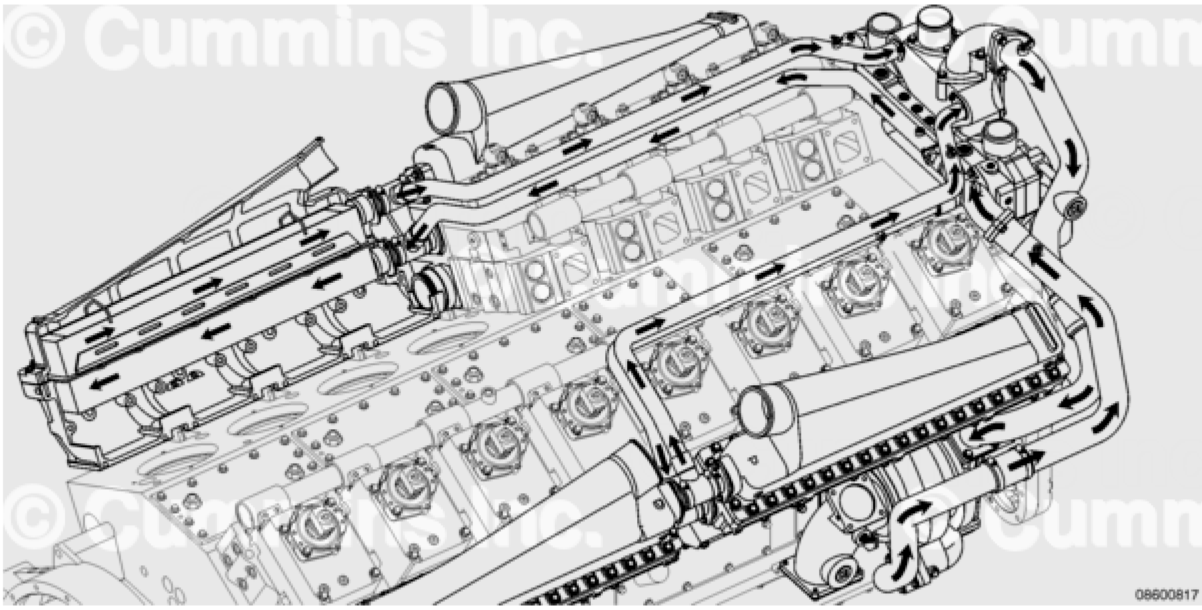

Figure 3, Coolant Cooling Loop

The QSK60 MCRS uses an aftercooler similar to the aftercooler used on the QSK60 mechanically actuated injector engine. An aftercooler is a coolant-to-air heat exchanger used to reduce the temperature of the air entering the intake manifold. Unlike the QSK60 mechanically actuated injector engine, the coolant that is supplied to the QSK60 MCRS aftercooler core is on a separate loop from the engine coolant. This coolant loop **must** have its own radiator. The aftercooler radiator **must** be designed to provide coolant cold enough to cool the intake air to 60°C [142 °F] on a 25°C [77°F] ambient day. The two cooling circuits share a common top tank. See Figure 4 below.

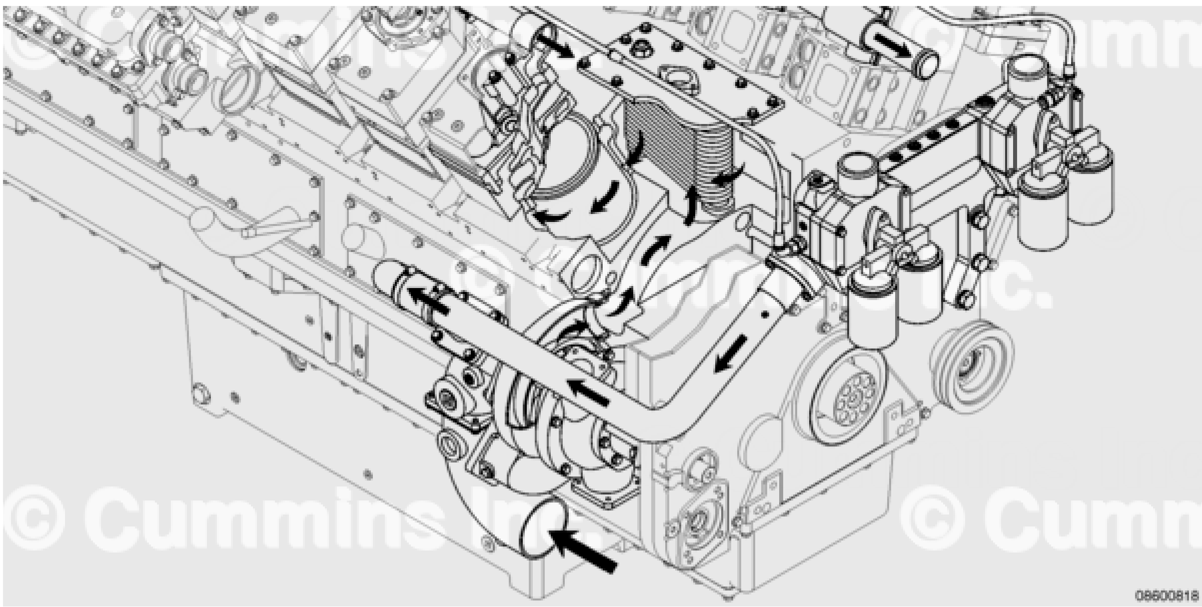

Figure 4, Aftercooler Coolant Loop

Last Modified: 10-May-2007